

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**  
**8<sup>th</sup> Semester of B. Pharm Examination**  
**University Theory Examination April 2015**  
**PH409 Herbal Drugs Technology-II**

**Date: 22/04/2015 , Wednesday    Time: 10:00 a.m. to 1:00 p.m.    Maximum Marks: 80**

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**Instructions:**

- 1. Figures to right indicate marks.**
- 2. Section wise separate answer book to be used.**
- 3. Draw neat sketches wherever necessary.**

**SECTION- I**

**Q.1    Attempt any TWO of the following:**

- |   |                                                                                                           |    |
|---|-----------------------------------------------------------------------------------------------------------|----|
| A | Give WHO Guidelines for standardization of crude drugs.                                                   | 10 |
| B | Define optical isomerism and Geometric isomerism giving examples of each.                                 | 10 |
| C | Explain Effect of stereochemistry of natural compounds on their medicinal value giving suitable examples. | 10 |

**Q.2    Attempt any TWO of the following:**

- |   |                                                                                                                                        |     |
|---|----------------------------------------------------------------------------------------------------------------------------------------|-----|
| A | Define plant tissue culture. Give a brief account on advantages of tissue culture techniques over the conventional culture techniques. | 10] |
| B | Enlist different types of tissue culture techniques. Give brief account on protoplast fusion and immobilization techniques.            | 10  |
| C | Write a note on nutritional requirement in plant tissue culture techniques.                                                            | 10  |

**SECTION-II**

**Q.3    Attempt any TWO of the following:**

- |   |                                                                               |    |
|---|-------------------------------------------------------------------------------|----|
| A | Give a note on applications of plant tissue culture with suitable examples.   | 10 |
| B | Write a note on silymarin and hypricin with respect its medicinal importance. | 10 |
| C | Give a detailed note on antiviral drugs derived from marine sources.          | 10 |

**Q.4 Attempt any TWO of the following:**

- A Enlist different types of chromatographic techniques. Give role of HPTLC in phytopharmaceutical evaluation. **10**
  - B Write a note on anticancer drugs obtain from marine sources. **10**
  - C Give different regulatory guideline prevailing world wide in herbal drugs trade. **10**
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